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(71)**Name of Applicant :**
**1)Pankaj Kunekar**
Address of Applicant :Vishwakarma Institute of Technology, Pune, Maharashtra, India Maharashtra India
**2)Sejal D'mello**
**3)Punit Ayare**
**4)Pankaj Bhandari**
**5)Devendra Gurav**

(72)**Name of Inventor :**
**1)Sejal D'mello**
**2)Punit Ayare**
**3)Pankaj Bhandari**
**4)Devendra Gurav**

(57) Abstract :
Falls among elderly individuals are a major health concern and often result in severe injuries, delayed medical assistance, and loss of independence. Many existing fall detection systems rely on manual triggers, simple threshold-based sensors, or intrusive camera-based monitoring, which limit their reliability, privacy, and accessibility. The present invention discloses an AI-enabled wearable system for real-time fall detection and automated caregiver alerting. The system is implemented as a lightweight wrist-worn device incorporating a motion sensing module comprising an accelerometer and gyroscope. The wearable device continuously monitors the user’s movement and orientation and transmits sensor data wirelessly to a cloud-based processing unit. An artificial intelligence model hosted on the cloud analyzes the incoming motion data in real time to accurately identify fall events while minimizing false alarms caused by normal daily activities. When a fall is detected with sufficient confidence, the system automatically logs the event and sends instant notifications to registered caregivers through a mobile application using cloud messaging services. The proposed system operates autonomously without requiring manual activation, ensuring reliability even when the user is unconscious or unable to seek help. The architecture emphasizes affordability, privacy preservation, and scalability, making it suitable for home care, assisted living facilities, and healthcare monitoring applications. By integrating wearable sensing, artificial intelligence, cloud computing, and mobile communication, the invention provides a comprehensive solution for improving elderly safety and emergency response effectiveness.

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